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Prevalence and associated factors of intrinsic capacity based on Andersen Model among older adults: a national cross-sectional study in China

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Abstract

Background Extensive evidence indicates intrinsic capacity (IC) as a strong predictor for health outcomes. Identifying conditions associated with IC impairment provides an opportunity to intervene to slow down, stop or reverse the declines. However, current evidence on factors affecting IC remains limited and inconsistent. This study aimed to identify factors associated with IC and the degree of IC impairment in older adults, thereby assisting in the development of comprehensive intervention strategies to mitigate the IC impairment.

Methods We analyzed data of 10,654 participants sourced from the fifth wave of the Chinese Longitudinal Healthy Longevity Survey (2020). Chi-square tests, and multinomial logistic regression analysis were employed to analyze the possible variables associated with IC impairment based on Andersen Model.

Results 10,617 participants were included into this study, and 86.9% of them have impairment in at least one dimension of IC. We identified 18 factors based on Andersen Model and 14 of them were associated with IC impairment. In the predisposing characteristics dimension of Andersen Model, older adults particularly those aged ≥ 80 years exhibited higher risk of IC impairment than those aged 60–69 years. Individuals living in urban areas, educated, drinking less than once a month, using the internet, and having social participation were less likely to experience IC impairment. In the enabling resources dimension of Andersen Model, individuals with access to pension, medical insurance, work, and spousal emotional support exhibited lower risk of IC impairment. In the need factors dimension of Andersen Model, individuals who self-rated their health as poor, and those with pain and multimorbidity exhibited higher risk of IC impairment. Individuals sleeping 5–7 h per night demonstrated a lower risk of IC impairment relative to more than 7 h.

Conclusion *More than half of the older adults exhibited varying degrees of IC impairment in China. Governments and community health systems should implement programs promoting rural urbanization, digital literacy, social*

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participation, and equitable allocation of medical insurance and pension resources to mitigate IC impairment in older adults.

Keywords Intrinsic capacity, Andersen Model, Associated factors, Healthy aging

Text box 1. Contributions to the literature

- This study identified factors associated with intrinsic capacity of older adults based on Andersen Model with data generated from Chinese Longitudinal Healthy Longevity Survey (2020), addressing the issue of single perspectives in existing research on this topic.
- Predisposing characteristics exemplified by age, living area, internet usage, enabling resources like working, pension, and medical insurance, and need factors including self-rated health, pain, comorbidities and nocturnal sleep duration were associated with intrinsic capacity impairment.
- Governments and community health systems should implement programs promoting rural urbanization, digital literacy, social participation, and equitable allocation of social resources to mitigate IC impairment in older adults.

Introduction

The demographic transition is one of the most challenging developments in our modern world [1]. In 2020, 9% of the world's population was 65 years and older, and by 2050, one in six people in the world will be over the age of 65 [2]. With the possibility of differences in individual aging rates and the constantly increasing lifespans, the idea of healthy aging as an optimal biological, sociological and physiological development throughout life arose [1]. The World Health Organization (WHO) published an action plan, "Decade of healthy aging 2020 ~ 2030", which calls for ten years of concerted, catalytic, sustained collaboration [3, 4]. And the WHO also outlines a model of healthy aging in 2015 with three primary factors: intrinsic capacity, functional ability, and environment [5]. Intrinsic capacity (IC) is the core of healthy aging, which is defined as "all the physical and mental capacities that an individual can draw on at any point" [5]. Five domains for measurements of IC were developed: locomotion, vitality, cognition, psychological and sensory capacities (including hearing and vision) [6]. In general, an individual's IC peaks in early adulthood and then gradually decreases as they get older [7]. Variation in IC is far greater across people in older age than across younger groups, and such diversity is one of the hallmarks of ageing [8].

The prevalence of IC impairment in older adults worldwide is 67.8% [9]. Though measurements of IC and derivation of the summative index score are heterogeneous [10, 11], extensive evidence indicates IC as a strong predictor for health outcomes, including disability, activities of daily living, recurrent falls, emergency department visits, quality of life, even mortality [12–14]. The WHO advocates for integrated care for older people (ICOPE) to prevent and mitigate the decline in IC, emphasizing that services should be delivered based on the needs of older adults rather than the necessity of the services themselves [15]. The Andersen Model [16, 17] addresses service delivery from a demand rather than supply-side perspective [18], aligning closely with WHO principles. It encompasses three dimensions: predisposing characteristics, enabling resources, and need factors. Based on the Andersen Model, it is possible to identify factors influencing healthcare resource utilization, which may affect IC in turn. This facilitates the development of comprehensive interventions to mitigate IC impairment.

China has the largest older population globally [19], making research on IC of older adults in this country particularly representative. Therefore, we conducted a study on the elderly population in China to assess the prevalence and identify factors associated with IC impairment based on Andersen Model. We hypothesized that factors such as predisposing characteristics, enabling resources, and need factors would be independently associated with IC impairment among older adults.

Methods

Theoretical framework

This study was underpinned by the Health Care Utilization Model originally propounded by Andersen and Newman [20], which was referred to as the "Andersen Model". There are three key elements in the Andersen Model: predisposing characteristics, enabling resources, and need factors [21]. Predisposing characteristics are primarily an individual's innate or acquired characteristics such as demographic, social structure, and health beliefs to use medical and health services before the onset of disease. Enabling resources refer to the resources possessed by individuals, families, and communities that enhance the likelihood of those in need seeking or utilizing relevant services. Need factors represents both self-perceived and evaluated need for health service [22–24].

Andersen Model is a multilevel theory primarily applied to services utilization studies across various fields such as sociology, medicine, public health, and psychology [25]. In recent years, this model has also been employed in research examining determinants of health-related issues to identify key variables, such as the factors influencing health literacy [26] and health poverty vulnerability [22]. In this study, Andersen Model presents a theoretical framework for a comprehensive understanding of the factors associated with IC impairment in older adults.

Data source and study population

Data were obtained from the China Health and Retirement Longitudinal Study (CHARLS), which is a large, nationally representative, prospective cohort of community-dwelling adults aged 45 years and above, covering 150 counties/districts and 450 villages/urban communities across 28 provinces of China [27]. Since the baseline of CHARLS was conducted in 2011, respondents were interviewed biennially on socioeconomic status and health circumstances via face-to-face computer-assisted personal interviews.

In this study, data were obtained from the fifth wave of the CHARLS (2020), with a total sample size of 19,395. We restricted our analysis to individuals aged ≥ 60 years (n = 10,654). Following the methodology of other studies, samples with > 20% missing data in key variables were excluded (n = 37) [28]. For remaining cases with < 20% missing values, continuous variables were imputed using mean substitution, while categorical variables were handled by mode imputation [28]. After these procedures, a final sample of 10,617 participants was included in the statistical analysis. Ethical approval and informed consent for CHARLS were received from the Peking University Ethical Review Committee.

Measures

Dependent variable

The dependent variable in this study was the level of IC, including five dimensions: cognition, locomotion, psychological, vitality, sensory capacities (including hearing and vision). Each dimension scored as 0 (impaired) or 1 (stable). The total score for IC was 0–5. We classified IC into three levels based on the score, with a score of 5 indicating high and stable IC, IC score ≥ 3 and < 5 as declining IC, and an IC score < 3 as significant loss of IC.

Assessment tools of IC

Cognitive capacity was assessed by the Mini-Mental State Examination (MMSE). MMSE has a total score of 0–30, with higher scores indicating better cognitive functioning. According to the educational attainment classification, people were defined to be cognitively impaired if they scored ≤ 17 in illiteracy, ≤ 20 in primary or below (literate), and ≤ 24 in junior high school and above [29]. The Cronbach's α coefficient for MMSE was 0.725 and the KMO value was 0.784.

Locomotor capacity was measured by the Long International Physical Activity Questionnaire (IPAQ). In the past 7 days, moderate-intensity physical activity >150 min/week, or higher-intensity physical activity ≥ 75 min/week, or the sum of moderate-intensity and higher-intensity physical activity >150 min/week was considered "adequate (unimpaired)" [30], otherwise would be judged as "inadequate (impaired)". The Cronbach's α coefficient for IPAQ was 0.723 and the KMO value was 0.799.

Psychological capacity was assessed by the Center for Epidemiologic Studies Depression Scale (CES-D). CES-D contains 10 entries with a total score of 0–30. A total score ≥ 20 was categorized as impaired psychological capacity. The Cronbach's α coefficient for CES-D was 0.844 and the KMO value was 0.909.

The total physical activity volume (PAV) was used to indicate vitality capacity. PAV can be expressed using the metabolic equivalent (MET) [31], which is calculated as follows: PAV = 8.0 × weekly vigorous physical activity duration score + 4.0 × weekly moderate physical activity duration score + 3.3 × weekly low intensity physical activity duration score [32]. According to the IPAQ, physical inactivity was indicated if the total PAV did not reach 600 MET-minutes/week, which was defined impaired vitality in this study.

Sensory ability was assessed from question DC029 of the CHARLS Wave 5 (2020) Follow-up Questionnaire, which went through the visual and hearing impairments observed by the interviewer. Sensory impairment was defined as impairment in either vision, hearing, or both.

Independent variables

We adapted Andersen Model for this study and examined a total of 18 factors related to IC across three dimensions (Fig. 1). The detailed definitions, coding schemes and sources for all independent variables are summarized in Table 1.

Statistical analysis

Statistical analyses were conducted using R (R Foundation for statistical computing, version 4.4.0) and SPSS software (version 26.0). Data were presented as n (%) for categorical variables and mean (SD) for continuous variables with normal distributions. Univariate analysis was performed using the chi-square test, and statistically significant variables were included in the multivariable logistic regression. We initially used ordinal logistic regression for multivariable analysis but failed the parallel-line test, then we switched to multinomial logistic regression. Significance level of $P < 0.10$ in univariate analysis and $P < 0.05$ in multinomial logistic regression were defined as statistically significant.

Previous literatures suggest that gender, age, living area, education attainment, work status, and social participation may exhibit complex interrelated effects [33–35]. To explore the intricate relationships among key social factors, we incorporated the following interaction terms into a multivariable logistic regression model: age × gender, educational attainment × living area, gender × living area, gender × working status, living area × working

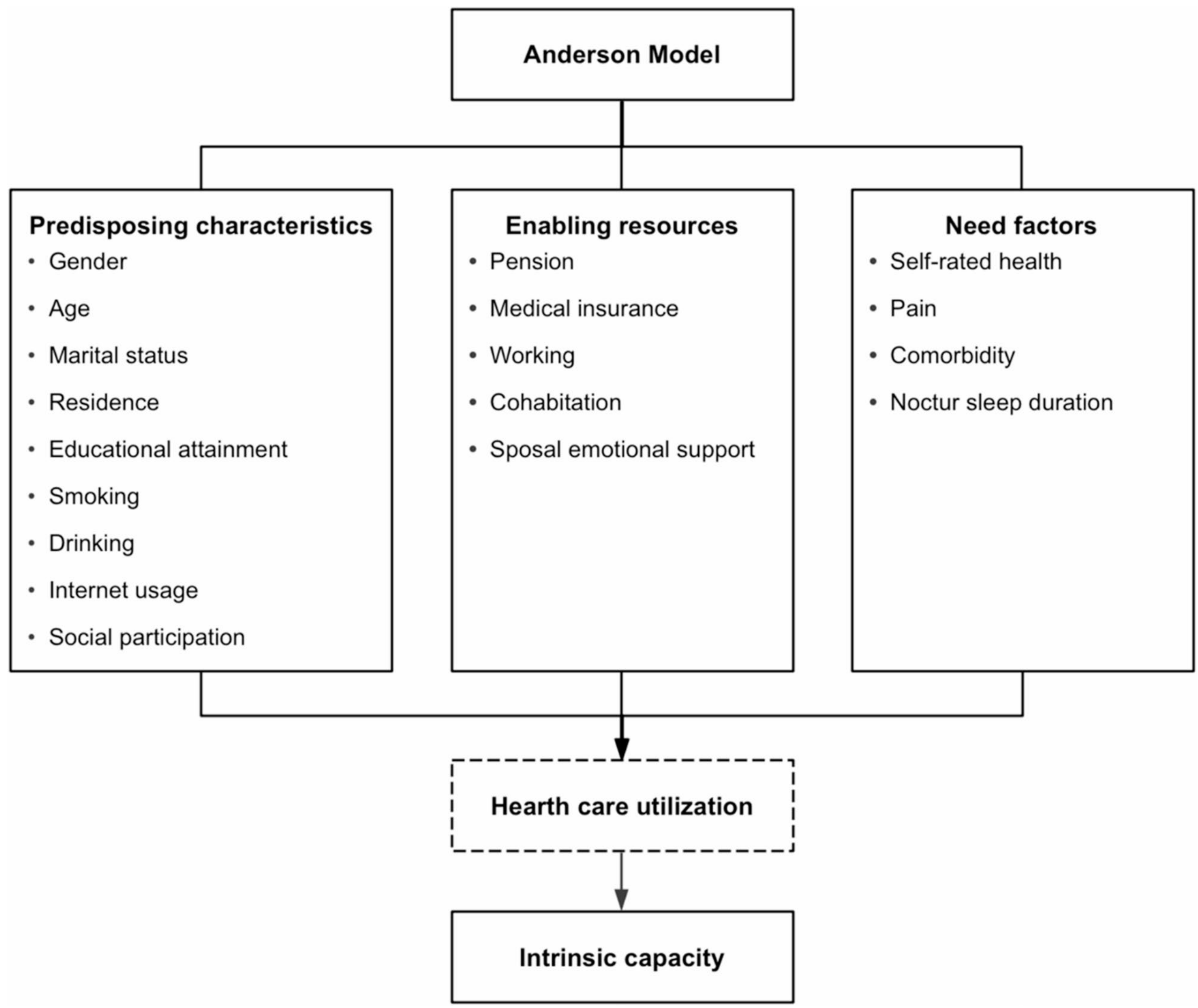


Fig. 1 Factors associated with IC based on adaptive Andersen Model

status, and social participation × working status. $P < 0.05$ indicates statistically significant differences.

Results

Intrinsic capacity of older adults

Of the 10,617 individuals, age ranged from 60 to 108 years, with a mean (SD) of 68.99 (6.90) years. The population demonstrated a mean (SD) IC score of 3.94 (0.69). Overall, 86.9% of participants exhibited declining or significant loss of IC, and 77.8% of them had 1 or 2 impaired capacities. The dimensions of IC impairment among older adults are shown in Table 2.

Factors associated with IC of older adults

Univariate analysis based on Andersen Model

The characteristics based on Andersen Model of 10,617 participants are shown in Table 3.. Seventeen of the 18 variables were associated with IC impairment of the elderly, except for the smoking factor ($P = 0.411$).

Multinomial logistic regression

The 17 variables that were statistically significant in the results of the univariate analysis were included in the multinomial logistic regression. Using high and stable IC group as a reference category, the results showed that, apart from gender, marital status and cohabitation status, the remaining 14 variables all exerted statistically significant effects on IC impairment.

The results of the multinomial logistic regression comparing the declining IC group with the high and stable IC group are shown in Fig. 2 (left). In the predisposing characteristics dimension of Andersen Model, older adults aged ≥ 80 years exhibited higher odds of IC decline than those aged 60–69 years (OR = 3.238, 95% CI 2.114–4.961, $P < 0.001$). Individuals with urban residence, primary

Table 1 Definitions and measurement of independent variables

Characteristics	Variable name	Response Options and Value Assignment	Question number corresponding to the questionnaire
Predisposing characteristics	Gender	Male = 1, Female = 2	BA001
	Age	60–69 years = 1, 70–79 years = 2, ≥ 80 years	BA003
	Marital status	Partnered[a] = 1, Non-partnered[b] = 2	BA011
	Living area	Urban = 1, Rural = 2	BA008
	Educational attainment	Illiteracy = 1, Primary or below (literate) = 2, Junior high school = 3	BA010
	Smoking status	Never smokers = 1, Former smokers = 2, Current smokers = 3	DA046, DA047
	Drinking	Non-drinkers = 1, Occasional drinkers[c] = 2, Regular drinkers[d] = 3	DA051
	Internet usage	No = 0, Yes = 1	DA040
	Social participation	No[e] = 0, Yes = 1	DA038
Enabling resources	Pension	No = 0, Yes = 1	GB006
	Medical insurance	No = 0, Yes = 1	BA016
	Working	No = 0, Yes = 1	FA001, FA004
	Cohabitation	No = 0, Yes = 1	BA018
	Spousal emotional support[f]	No[f] = 0, Yes[g] = 1	BA011
Need factors	Self-rated health	Very poor = 1, Poor = 2, Fair = 3, Good = 4, Very good = 5	DA001
	Pain	No pain = 1, Slight pain = 2, Moderate pain = 3, Severe pain = 4, Very severe pain = 5	DA027
	Comorbidity	None = 0, Single = 1, Multimorbidity = 2	DA003
	Nocturnal sleep duration	≤ 5 h = 1, > 5 but ≤ 7 h = 2, > 7 h = 3	DA030

[a] Currently married or cohabiting

[b] Never married, separated, divorced, or widowed

[c] Alcohol consumption < 1 drinking occasion per month

[d] Alcohol consumption ≥ 1 drinking occasion per month

[e] Never participated in any social activities in the past month

[f] Divorced, widowed, unmarried, or separated for some reason

[g] Married or living with spouse

Table 2 Condition of IC impairment of the older adults ($n = 10{,}617$)

Variable	Category	*n* (%)	Mean IC Score (SD)
Intrinsic capacity	High and stable capacity (IC = 5)	1390 (13.1)	5.00 ± 0.00
	Declining capacity (3 ≤ IC<5)	8258 (77.8)	3.93 ± 0.44
	Significant loss of capacity (IC<3)	969 (9.1)	2.62 ± 0.46
Cognitive capacity	Impaired	6790 (64.0)	
	Unimpaired	3827 (36.0)	
Locomotor capacity	Impaired	634 (6.0)	
	Unimpaired	9983 (94.0)	
Psychological capacity	Impaired	743 (7.0)	
	Unimpaired	9874 (93.0)	
Vitality	Impaired	471 (4.4)	
	Unimpaired	10,146 (95.6)	
Sensory capacity	Impaired	940 (8.9)	
	Unimpaired	9677 (91.1)	

school education or below, alcohol consumption of less than once per month, internet access, or social participation had a lower risk of IC decline (all OR < 1, $P < 0.05$), compared to those with rural residence, illiteracy, no alcohol consumption, lack of internet access, or no social participation. There was no statistically significant impact of gender and marital status on IC ($P > 0.05$). Within the enabling resources dimension of Andersen Model, individuals with pensions, medical insurance, work, or spousal emotional support had a lower risk of IC decline (all OR < 1, $P < 0.05$), compared to those without pensions, medical insurance, work, or spousal emotional support. No such association was observed for cohabitation status ($P > 0.05$). Within the need factors dimension of Andersen Model, individuals with fair or poor self-rated health, multimorbidity had a higher risk of IC decline (OR > 1, $P < 0.05$), compared to those with very good self-rated health or no comorbidities. Conversely, those with moderate pain or a sleep duration of 5–7 h per night had a lower risk (OR < 1, $P < 0.05$) compared to those with no pain or a sleep duration of > 7 h.

The results of the multinomial logistic regression comparing the significant loss of IC group with the high and stable IC group are shown in Fig. 2 (right). In the predisposing characteristics dimension of Andersen Model, older adults aged 70–79 years and ≥ 80 years exhibited higher odds of significant loss of IC than those aged 60–69 years (OR > 1, $P < 0.001$). Individuals with urban

Table 3. Univariate analysis of factors associated with IC based on Andersen Model in older adults ($n = 10,617$)

Characteristic	Variable	IC (n, %)			Total (n, %)	χ^2	*P* value
		High and stable capacity	Declining capacity	Significant loss of capacity			
Predisposing characteristics	Gender					44.901	<0.001*
	Male	785 (15.3)	3923 (76.3)	431 (8.4)	5139 (100%)		
	Female	605 (11.0)	4335 (79.2)	538 (9.8)	5478 (100%)		
	Age					293.912	<0.001*
	60–69 years	1018 (15.7)	5057 (78.0)	412 (6.3)	6487 (100%)		
	70–79 years	348 (11.1)	2408 (76.7)	384 (12.2)	3140 (100%)		
	≥ 80 years	24 (2.4)	793 (80.1)	173 (17.5)	990 (100%)		
	Marital status					156.913	<0.001*
	Partnered	1207 (14.9)	6260 (77.3)	628 (7.8)	8095 (100%)		
	Non-partnered	183 (7.3)	1998 (79.2)	341 (13.5)	2522 (100%)		
	Living area					75.579	<0.001*
	Rural	798 (11.4)	5475 (78.3)	720 (10.3)	6993 (100%)		
	Urban	592 (16.3)	2783 (76.8)	249 (6.9)	3624 (100%)		
	Educational attainment					197.774	<0.001*
	Illiteracy	232 (7.1)	2628 (80.8)	395 (12.1)	3255 (100%)		
	Primary or below (literate)	733 (15.8)	3503 (75.3)	413 (8.9)	4649 (100%)		
	Junior high school or above	425 (15.7)	2127 (78.4)	161 (5.9)	2713 (100%)		
	Smoking status					3.961	0.411
	Never smokers	794 (12.7)	4892 (78.1)	575 (9.2)	6261 (100%)		
	Former smokers	227 (13.0)	1349 (77.5)	166 (9.5)	1742 (100%)		
	Current smokers	369 (14.1)	2017 (77.2)	228 (8.7)	2614 (100%)		
	Drinking					103.133	<0.001*
	Non-drinkers	800 (11.1)	5653 (78.6)	742 (10.3)	7195 (100%)		
	Occasional drinkers	155 (18.0)	651 (75.5)	56 (6.5)	862 (100%)		
	Regular drinkers	435 (17.0)	1954 (76.3)	171 (6.7)	2560 (100%)		
	Internet usage					297.600	<0.001*
	No	873 (10.5)	6603 (79.0)	877 (10.5)	8353 (100%)		
	Yes	517 (22.8)	1655 (73.1)	92 (4.1)	2264 (100%)		
	Social participation					114.793	<0.001*
	No	621 (10.4)	4711 (78.9)	639 (10.7)	5971 (100%)		
	Yes	769 (16.6)	3547 (76.3)	330 (7.1)	4646 (100%)		
Enabling resources	Pension					22.229	<0.001*
	No	210 (10.1)	1668 (79.8)	210 (10.1)	2088 (100%)		
	Yes	1180 (13.8)	6590 (77.3)	759 (8.9)	8529 (100%)		
	Medical insurance					41.904	<0.001*
	No	35 (6.1)	452 (79.3)	83 (14.6)	570 (100%)		
	Yes	1355 (13.5)	7806 (77.7)	886 (8.8)	10,047 (100%)		
	Working					243.538	<0.001*
	No	499 (9.6)	4018 (77.6)	664 (12.8)	5181 (100%)		
	Yes	891 (16.4)	4240 (78.0)	305 (5.6)	5436 (100%)		
	Cohabitation					24.808	<0.001*
	No	271 (11.8)	1749 (76.5)	267 (11.7)	2287 (100%)		
	Yes	1119 (13.4)	6509 (78.2)	702 (8.4)	8330 (100%)		
	Spousal emotional support					122.199	<0.001*
	No	247 (8.3)	2357 (79.3)	369 (12.4)	2973 (100%)		
	Yes	1143 (14.9)	5901 (77.2)	600 (7.9)	7644 (100%)		

Table 3 (continued)

Characteristic	Variable	IC (n, %)			Total (n, %)	χ^2	*P* value
		High and stable capacity	**Declining capacity**	**Significant loss of capacity**			
Need factors	Self-rated health					275.711	<0.001*
	Very good	167 (17.7)	717 (76.2)	57 (6.1)	941 (100%)		
	Good	181 (16.8)	824 (76.7)	70 (6.5)	1075 (100%)		
	Fair	794 (13.5)	4678 (79.6)	406 (6.9)	5878 (100%)		
	Poor	191 (9.5)	1525 (75.9)	294 (14.6)	2010 (100%)		
	Very poor	57 (8.0)	514 (72.1)	142 (19.9)	713 (100%)		
	Pain					206.993	<0.001*
	No pain	600 (14.0)	3379 (78.6)	317 (7.4)	4296 (100%)		
	Slight pain	436 (14.7)	2315 (78.1)	214 (7.2)	2965 (100%)		
	Moderate pain	170 (15.9)	792 (74.2)	105 (9.9)	1067 (100%)		
	Severe pain	100 (9.4)	852 (80.3)	109 (10.3)	1061 (100%)		
	Very severe pain	84 (6.8)	920 (74.9)	224 (18.3)	1228 (100%)		
	Comorbidity					25.994	<0.001*
	None	1030 (13.9)	5699 (77.2)	658 (8.9)	7387 (100%)		
	Single	264 (12.0)	1746 (79.3)	192 (8.7)	2202 (100%)		
	Multimorbidity	96 (9.3)	813 (79.1)	119 (11.6)	1028 (100%)		
	Nocturnal sleep duration					134.551	<0.001*
	>7 h	283 (11.2)	2036 (80.5)	209 (8.3)	2528(100%)		
	>5, ≤7 h	654 (17.3)	2861 (75.8)	262 (6.9)	3777 (100%)		
	≤5 h	453 (10.5)	3361 (78.0)	498 (11.5)	4312 (100%)		

*$P < 0.001$

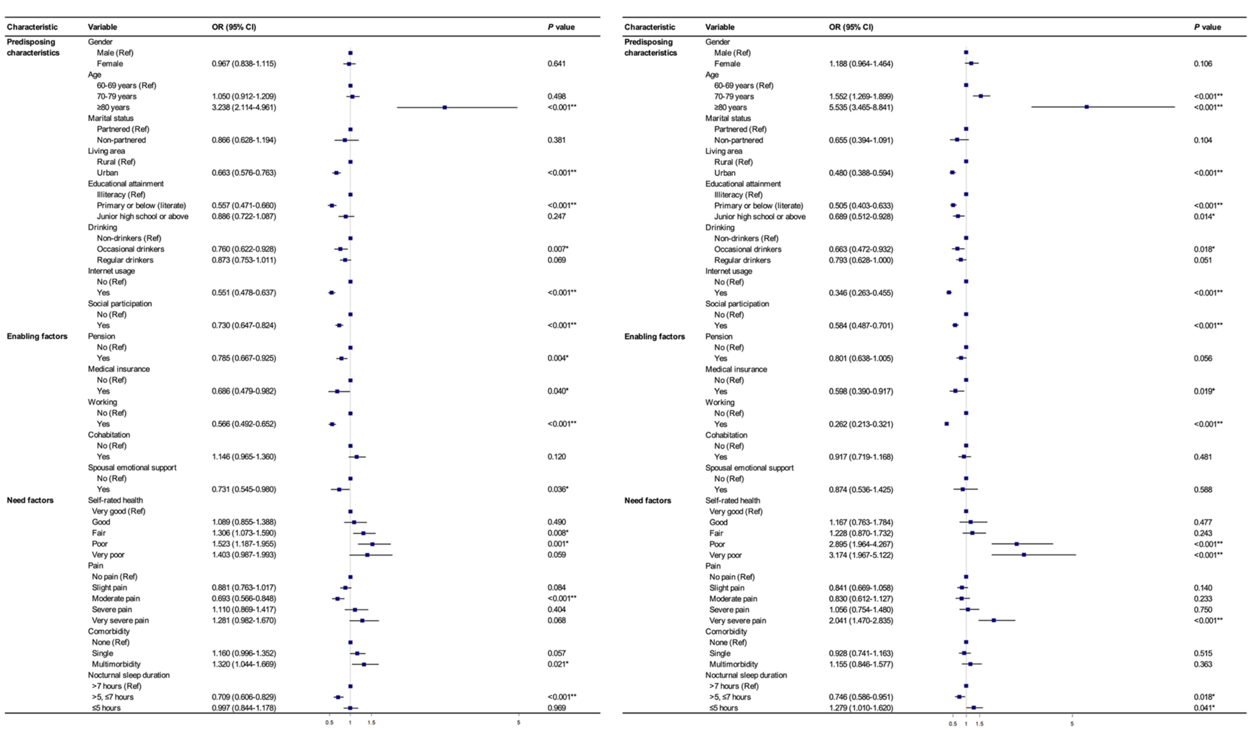
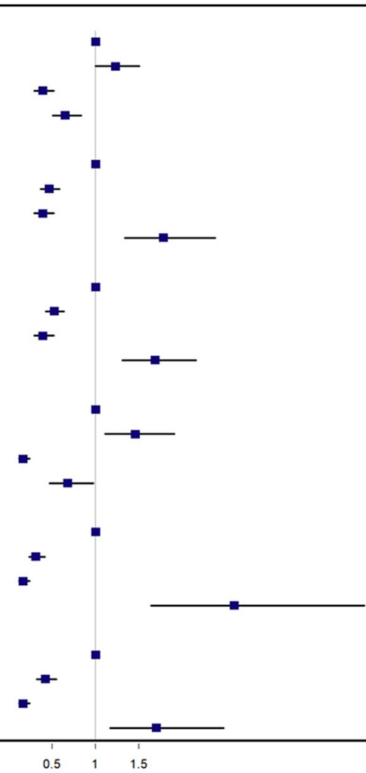


Fig. 2 Multinomial logistic regression analysis of factors associated with IC based on Andersen Model in older adults (using high and stable IC group as reference)

Intrinsic capacity	Interaction terms	OR (95% CI)	*P for interaction*
Declining	Gender × Working		
	Female × Not working (Ref)		
	Male × Not working	1.227 (0.997-1.510)	0.054
	Female × Working	0.390 (0.295-0.517)	0.000**
	Male × Working	0.650 (0.506-0.834)	0.001*
	Living area × Working		
	Rural × Not working (Ref)		
	Urban × Not working	0.465 (0.369-0.586)	0.000**
	Rural × Working	0.390 (0.295-0.517)	0.000**
	Urban × Working	1.786 (1.340-2.381)	0.000**
	Social participation × Working		
	No participation × Not working (Ref)		0.000**
	Participation × Not working	0.523 (0.427-0.641)	0.000**
	No participation × Working	0.390 (0.295-0.517)	0.000**
	Participation × Working	1.684 (1.311-2.163)	
Significant loss	Gender × Working		
	Female × Not working (Ref)		
	Male × Not working	1.459 (1.112-1.914)	0.006*
	Female × Working	0.166 (0.116-0.238)	0.000**
	Male × Working	0.677 (0.468-0.980)	0.039*
	Living area × Working		
	Rural × Not working (Ref)		
	Urban × Not working	0.313 (0.235-0.418)	0.000**
	Rural × Working	0.166 (0.116-0.238)	0.000**
	Urban × Working	2.596 (1.642-4.103)	0.000**
	Social participation × Working		
	No participation × Not working (Ref)		
	Participation × Not working	0.421 (0.324-0.547)	0.000**
	No participation × Working	0.166 (0.116-0.238)	0.000**
	Participation × Working	1.698 (1.165-2.475)	0.006*



* *P*<0.05; ** *P*<0.001

Fig. 3 Interaction effects of factors associated with IC of older adults

residence, education, alcohol consumption of less than once per month, internet access, or social participation had a lower risk of significant loss of IC (all OR < 1, $P < 0.05$), compared to those with rural residence, illiteracy, no alcohol consumption, lack of internet access, or no social participation. There was no statistically significant impact of gender and marital status on IC ($P > 0.05$). Within the enabling resources dimension of Andersen Model, individuals with medical insurance and work had a lower risk of significant loss of IC (all OR < 1, $P < 0.05$), compared to those without medical insurance and work. There were no statistically significant impact of pension, cohabitation, and spousal emotional support on significant loss of IC (all $P > 0.05$). Within the need factors dimension of Andersen Model, individuals with poor or very poor self-rated health and very severe pain were associated with a higher risk of significant loss of IC (OR > 1, $P < 0.001$), compared to those with very good self-rated health and no pain. Compared with older adults who sleep more than 7 h per night, those who consistently sleep between 5 and 7 h per night had a lower risk of significant loss of IC (OR = 0.746, 95% CI 0.586–0.951, $P = 0.018$). Conversely, individuals who sleep less than 5 h per night faced a higher risk of significant loss of IC (OR = 1.279, 95% CI 1.010–1.620, $P = 0.041$).

Interaction effects

We incorporated 6 predefined interaction terms into multinomial logistic regression model to explore whether synergistic or antagonistic effects exist among these key social factors. Compared with the high and stable IC group, statistically significant interactions were observed in both the declining IC group and significant loss of IC group. Our analysis revealed significant interaction effects for gender × working status, living area × working status, and social participation × working status (Fig. 3). But no interactions were observed for the effects of age × gender, educational attainment × living area, or gender × living area on IC.

In both groups, regardless of gender, working individuals had a lower risk of IC impairment compared to those without work (OR < 1, $P < 0.05$). However, when there was not working, males were more likely than females to experience significant loss of IC (OR = 1.459, 95% CI 1.112–1.914, $P = 0.006$). Compared to individuals living in rural areas who were not working, those residing in urban areas without work or those in rural areas with work had a lower risk of developing IC impairment (OR < 1, $P < 0.001$). Conversely, individuals living in urban areas with work faced an increased risk of IC impairment (OR > 1, $P < 0.001$). Compared to individuals who were socially inactive and not working, those who had social participation but were not working or socially inactive but working had a lower risk of IC impairment (OR < 1, P < 0.001). Whereas individuals who had social participation and work were more likely to experience IC impairment (OR > 1, P < 0.05).

Discussion

This study was a national cross-sectional survey that investigated the prevalence of IC impairment in 10,617 older adults in China. We found that 86.9% of the participants exhibited impairment in at least one dimension of IC. This rate was higher than the global pooled prevalence of 67.8% reported in prior systematic reviews [9]. Existing studies have demonstrated that the prevalence of IC impairment among older adults in China varies between 17.1% and 98.2% [9, 36], with a pooled estimate of 73.7% based on meta-analyses [37]. These differences might be due to disparities in sampling method and measurement instruments, as there is no consensus on a standardized scoring system for the screening tool [38]. Another possible reason is the large difference in sample size between the different studies. Our findings highlight the high prevalence of IC impairment among older adults, underscoring the urgent need for evidence-based interventions. Future studies should focus on standardizing IC assessment protocols and developing targeted intervention strategies.

Based on Andersen Model, this study identified 14 factors associated with IC impairment among older adults across three dimensions of predisposing characteristics, enabling resources and need factors.

In the predisposing characteristics dimension of Andersen Model, our study found that individuals with advanced age tended to have a more severe degree of IC impairment, which was especially pronounced among participants at ≥ 80 years of age. The association between age and IC was consistent with previous studies, which found that people in early old age were more likely to have better IC [36, 39]. Other predisposing characteristics such as living in urban areas, receiving education, using the internet, and having social participation all reduced the risk of IC impairment to varying degrees. Possible reasons might be that individuals possessing these characteristics tend to gain and accumulate greater contextual and psychosocial support, which in turn maintains their IC. Our findings suggested that drinking less than once a month may help reduce the risk of IC impairment. We postulated that this protective effect may be primarily attributed to the mood-enhancing benefits of alcohol, rather than to its protective effect on cognitive function. A previous 30-year (1985–2015) observational cohort study indicated that small amounts of alcohol did not confer protective effects on cognitive function [40]. These findings suggest that we need to pay close attention to the IC impairment of older adults, particularly those of advanced age. Interventions that target the acquired characteristics—such as promoting the urbanization of rural areas, providing health education, enhancing digital literacy, and fostering social participation—can thereby lead to mitigate IC impairment in older adults.

In the enabling resources dimension of Andersen Model, individuals with access to medical insurance, pensions, working, and spousal emotional support exhibited a lower risk of IC impairment. Medical insurance plays a significant role in material security and economic security for the elderly, which facilitates the function of spiritual security and subjective comfort and enhances their sense of security for their future life [41]. Retirement pension, working and spousal emotional support are mainly used to ensure the economic and emotional needs of older adults [39]. Therefore, it is necessary to expand the coverage of pension and medical insurance programs and provide emotional support to mitigate IC impairment of the elderly.

In the need factors dimension of Andersen Model, our study showed that older people with poor self-rated health, those experienced very severe pain, and those with comorbidities were more likely to have IC impairment than those who did not perceive such issues. These findings were consistent with previous study that older adults with good self-rated health were more likely to be classified as the stable and high IC trajectory group [39]. We found that, compared with those who slept > 7 h per night, people slept > 5 h but ≤ 7 h were less likely to have IC impairment, but those who slept ≤ 5 h tended to have more significant loss of IC. This finding was similar to previous study, which have shown a u-shaped relationship between IC and sleep duration [42]. These findings suggested that enhancing pain and comorbidity management, while encouraging healthy nocturnal sleep duration of 5–7 h, can help mitigate IC impairment of older adults.

All statistically significant interaction terms in this study involved working, suggesting a robust association between working and IC. Both men and women with working were associated with lower risk of IC impairment. However, among those not working, men faced a higher risk of significant loss of IC than women, indicating a pronounced gender disparity in this subgroup. This may be because men are generally more inclined to work to support their families and only cease working when they experience a significant loss of IC. Our study identified a lower risk of IC impairment among those residing in urban areas without work or those in rural areas with work as compared with individuals living in rural areas who were not working. This indicated that living in urban area or working each possesses an independent protective effect on IC. However, it is noteworthy that individuals who were both residing in urban areas and working faced an increased risk of IC impairment. Generally speaking, older adults in urban areas tend to be in a non-working, retired state. However, if they are still working, it is often due to economic necessity, which means they face greater financial and psychological pressures, thereby increasing the risk of IC impairment. Our study identified a lower risk of IC among those who had social participation but were not working or who were socially inactive but working as compared with individuals who were socially inactive and not working. Similarly, this indicated that social participation or working each possesses an independent protective effect on IC. But it is unclear why older adults who engaged in social activities while working were more likely to experience significant loss of IC, and more research is needed to explain this pattern. But this suggested that governments and community health systems should consider giving more attention to these two elderly groups, in order to mitigate their IC impairment.

This study had several limitations. First, although this study was based on a large national sample, its cross-sectional design limits causal inference. Consequently, we could only identify factors associated with IC rather than determine causal relationships. Additionally, it could not avoid recall bias associated with self-reported measures in questionnaire surveys. Second, previous studies showed that the physical environment plays a pivotal role in shaping human health behaviors and outcomes [43, 44]. However, CHARLS is not a specialized survey targeting the population's IC, and thus lacks the key confounders such as diet, quality of physical activity, and physical environment, which may introduce bias into the findings. Third, we imputed missing data by mean/mode imputation, which may introduce potential bias.

Conclusion

This study indicated that more than half of the older adults exhibited varying degrees of IC impairment in China. Based on the Andersen Model, factors spanning its three core dimensions—namely predisposing characteristics exemplified by age, living area, internet usage; enabling resources such as working, pension, and medical insurance; and need factors including self-rated health, pain, comorbidities and so on—were associated with IC impairment. We recommend that governments and community health systems implement programs promoting rural urbanization, digital literacy, social engagement, and equitable allocation of medical insurance and pension resources to mitigate IC impairment in older adults.

Abbreviations

CES-D	Center for Epidemiologic Studies Depression Scale
CHARLS	China Health and Retirement Longitudinal Study
IC	Intrinsic Capacity
IPAQ	International Physical Activity Questionnaire
IQ	Intelligence Quotient
MET	Metabolic Equivalent
MMSE	Mini-Mental State Examination
PAV	Physical Activity Volume
WHO	World Health Organization

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Authors' contributions
Guanyang Zou, Wenbin Zou, and Fengling Lu conceptualized and designed the study. Wenbin Zou and Fengling Lu collected and analyzed the data. Fengling Lu and Guanyang Zou drafted the paper, while Xiaolin Wei, Wenbin Zou, Yi Li and Wei Wei reviewed the paper and provided critical comments. All authors approved the final version.

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Data availability
Data are available in a public, open access repository. All the data are accessible to researchers around the world at the CHARLS project website: http://charls.pku.edu.cn.

Declarations

Ethics approval and consent to participate
Ethical approval for all the CHARLS waves was granted from the Institutional Review Board at Peking University. The IRB approval number for the main household survey, including anthropometrics, is IRB00001052-11015; the IRB approval number for biomarker collection was IRB00001052-11014. During the fieldwork, each respondent who agreed to participate in the survey was asked to sign two copies of the informed consent. Participants gave informed consent to participate in the study before taking part. The study conforms with the principles outlined in the Declaration of Helsinki.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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